Aryabhata conceived of the elliptical orbits of the planets a thousand years before Kepler, in the West, came to the same conclusion (having assumed, like all Europeans, that planetary orbits were circular rather than elliptical). He even estimated the value of the year at 365 days, six hours, 12 minutes and 30 seconds; in this he was only a few minutes off (the correct figure is just under 365 days and six hours). The translation of the *Aryabhatiya* into Latin in the thirteenth century taught Europeans a great deal; it also revealed to them that an Indian had known things that Europe would only learn of a millennium later.

The Vedic civilisation subscribed to the idea of a spherical earth at a time when everyone else, even the Greeks, assumed the earth was flat. By the fifth century CE Indians had calculated that the age of the earth was 4.3 billion years; as late as the nineteenth century, English scientists believed the earth was a hundred million years old, and it is only in the late twentieth century that Western scientists have come to estimate the earth to be about 4.6 billion years old.

India invented modern numerals (known to the world as 'Arabic' numerals because the West got them from the Arabs, who first encountered them in India). It was an Indian who first conceived of the zero, *shunya*; the concept of nothingness, *shunyata*, integral to Hindu and Buddhist thinking, simply did not exist in the West. Modern mathematics would have been impossible without the zero and the decimal system; just read a string of Roman numbers, which had no zeros, to understand their limitations.

Indian mathematicians invented negative numbers as well. The concept of infinite sets of rational numbers was understood by Jain thinkers in the sixth century BCE. Our forefathers can take credit for geometry, trigonometry, and calculus; the 'Bakhshali manuscript', seventy leaves of bark dating back to the early centuries of the Christian era, reveals fractions, simultaneous equations, quadratic equations, geometric progressions and even calculations of profit and loss, with interest.

The *Sulba Sutras*, composed between 800 and 500 BCE, demonstrate that India had Pythagoras's theorem before the great Greek was born, and a way of getting the square root of 2 correct to five decimal places.